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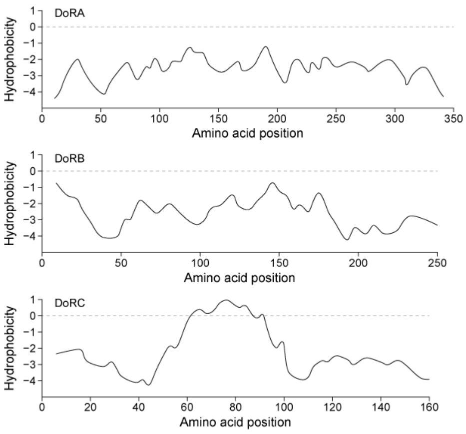

Figure 1 Hydrophobicity plots of three reflectins isolated from *D. opalescens*. (Note: Negative numbers indicate hydrophilic regions.)

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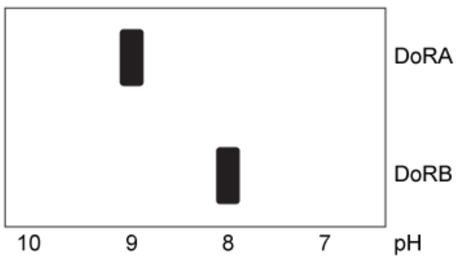


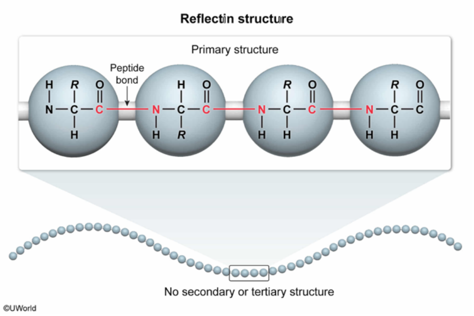
Figure 2 Isoelectric focusing of DoRA and DoRB

In the absence of ACh, which of the following bond types best characterizes the structure of reflectins?

- A. Hydrogen bonds between backbone carbonyls and amides [21%]
- B. Covalent bonds between sulfur atoms [3%]
- C. Hydrogen bonds between R group alcohols [24%]
- ✔ D. Covalent bonds between carbon and nitrogen atoms [50%]

Correct
51% answered correctly

Explanation:



Protein structure is organized in four levels as follows:

- **Primary structure** is the sequence of amino acids in a polypeptide chain. The amino acids are linked together by **covalent bonds** between the carbonyl carbon of one amino acid and the backbone nitrogen of another, called **peptide bonds**.

- **Secondary structure** is the set of local folds within a polypeptide and usually consists of **α-helices** and **β-sheets**. These structures are primarily stabilized by **hydrogen bonds** between the carbonyls and amides of the polypeptide **backbone**. The passage states that reflectins have few defined secondary structures, so hydrogen bonds between backbone carbonyls and amides are rare in reflectins **(Choice A)**.

- **Tertiary structure** is the overall three-dimensional structure of the protein and is stabilized by interactions between R groups, including hydrogen bonding, ionic, and hydrophobic interactions as well as disulfide bonds. The passage states that reflectins do not have well-defined tertiary structures, and therefore hydrogen bonds between R group alcohols are rare **(Choice C)**. The passage also states that reflectins have few cysteine residues, so covalent bonds between sulfur atoms (disulfide bonds) are also rare **(Choice B)**.

- **Quaternary structure** is the interaction of **two or more polypeptide chains** with each other and is stabilized by the same forces as tertiary structure. The aggregation of reflectins may be thought of as a type of quaternary structure, but the question asks for bond types in the absence of ACh, in which aggregation does not occur.

Because reflectins do not fold, they only participate in primary structure, characterized by the **covalent**

carbon-nitrogen linkages of peptide bonds.

Educational objective:
Protein structure is organized into primary, secondary, tertiary, and quaternary levels. Primary structure is the sequence of amino acids linked together by peptide bonds. Secondary structure consists of local structures such as α-helices and β-sheets stabilized by backbone hydrogen bonds. Tertiary and quaternary structures describe the three-dimensional form of a protein, mediated by hydrophobic, ionic, and hydrogen bond interactions.

Time spent:0 sec	**QID**:402379
Subject: Biochemistry	**System:** Amino Acids and Proteins

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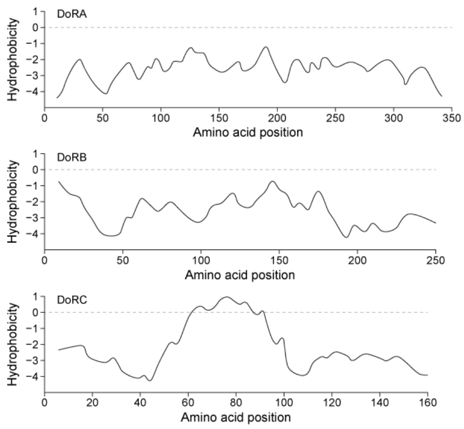

Figure 1 Hydrophobicity plots of three reflectins isolated from *D. opalescens*. (Note: Negative numbers indicate hydrophilic regions.)

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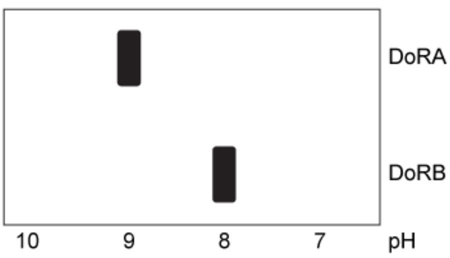


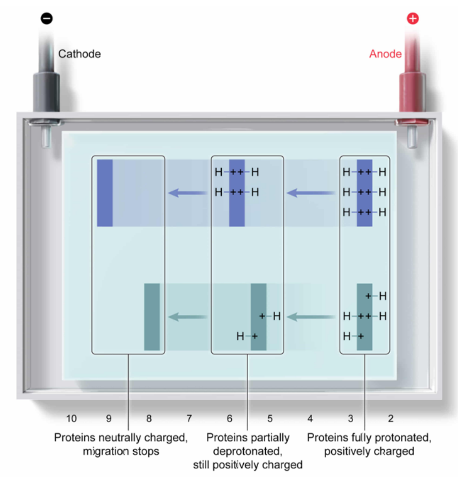
Figure 2 Isoelectric focusing of DoRA and DoRB

Suppose that to generate Figure 2, researchers loaded reflectins onto the low-pH side of a gel with a pH gradient, and a current was applied. This would result in:

- ✔ A. gradual deprotonation as reflectins migrated from anode to cathode. [39%]
- B. gradual deprotonation as reflectins migrated from cathode to anode. [31%]
- C. gradual protonation as reflectins migrated from anode to cathode. [18%]
- D. gradual protonation as reflectins migrated from cathode to anode. [10%]

Correct
39% answered correctly

Explanation:



In isoelectric focusing, an electric field causes proteins to migrate through a **pH gradient** within a gel (ie, one end of the gel has a low pH, and the pH within the gel gradually increases toward the opposite end, which has a high pH). Isoelectric focusing occurs in an electrolytic cell, in which the anode is positively charged and attracts anions, whereas the cathode is negatively charged and attracts cations. Note that these charges are the opposite of what they would be in a galvanic cell.

The question states that the proteins are loaded at the low-pH end of the gel. At **low pH**, proteins are fully **protonated and positively charged**, causing them to migrate through the gel **toward the negatively charged cathode**. Note that this is the opposite of what occurs in many techniques in which negatively charged particles migrate toward the positively charged anode (eg, SDS-PAGE or isoelectric focusing with proteins loaded at the *high*-pH end).

As positively charged proteins move toward the cathode, the **pH of the gel** that they are migrating in **gradually increases**, causing the proteins to become deprotonated and lose positive charge. Eventually, the **net charge becomes zero** and the protein **stops migrating**. The pH at which this occurs is the protein's isoelectric point. Therefore, in the given scenario, reflectins gradually become deprotonated as they move from anode to cathode until they reach their pI.

(Choice B) The low-pH end of the gel is near the *anode*, and proteins that start at this end migrate toward the *cathode*.
(Choice C) Reflectins migrate from anode to cathode in this scenario, but they become *de*protonated as they do so.

(Choice D) Reflectins would become protonated as they migrate from cathode to anode if they started at the *high*-pH side, but not at the low-pH side.

Educational objective:
In isoelectric focusing, an electric field causes proteins to migrate through a pH gradient. The low-pH end of the gradient is placed near the anode and the high-pH end is placed near the cathode. As proteins migrate from low pH to high pH (from anode to cathode), they lose protons and become less positively charged, stopping when their net charge becomes zero. The opposite is true of proteins migrating from high pH to low pH.

Time spent:0 sec
Subject:
Biochemistry

QID:402380
System:
Amino Acids and Proteins

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Skin cells from the squid *Doryteuthis opalescens* were cultured and three reflectins—DoRA, DoRB, and DoRC—were purified and analyzed to determine amino acid composition. They were found to contain an abundance of arginine, serine, and aspartate residues but relatively few cysteine, leucine, isoleucine, valine, phenylalanine, or tryptophan residues. Figure 1 shows the relative hydrophobicity of each reflectin variant as a function of amino acid position.

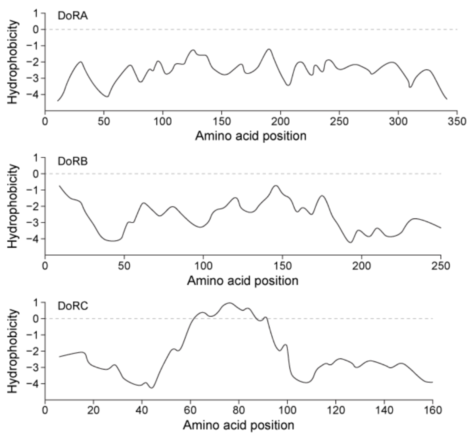


Figure 1 Hydrophobicity plots of three reflectins isolated from *D. opalescens*. (Note: Negative numbers indicate hydrophilic regions.)

In *D. opalescens* skin cells (pH ≈ 7.8), reflectins aggregate in response to acetylcholine (ACh), which causes a signal cascade that changes the phosphorylation state of the proteins. Cells were incubated with or without ACh, and DoRA and DoRB were purified and assessed for phosphorylation by mass spectrometry. DoRA became phosphorylated in the presence of ACh whereas DoRB became dephosphorylated. After examining the isoelectric points of unmodified DoRA and DoRB, as shown in Figure 2, scientists proposed that reflectins aggregate upon charge neutralization.

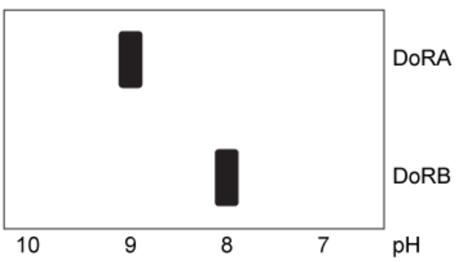


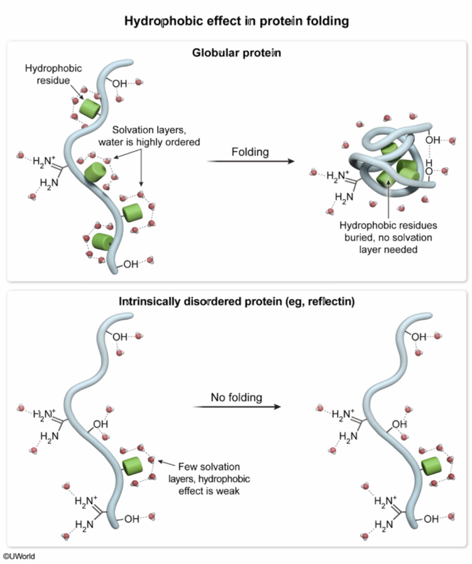
Figure 2 Isoelectric focusing of DoRA and DoRB

Which reason best explains why reflectin proteins do not adopt a defined tertiary structure?

- A. Reflectins are unable to interact with water efficiently. [5%]
- ✔ B. Reflectins only experience a weak hydrophobic effect. [72%]
- C. Reflectins cannot participate in hydrogen bonding. [14%]
- D. Reflectins are highly basic and therefore cannot fold. [7%]

Correct
73% answered correctly

Explanation:



Tertiary structure is the three-dimensional **folded form** of a protein. It is stabilized by multiple forces, including hydrogen bonding and ionic interactions between side chains, but the **primary driving force** in protein folding is the **hydrophobic effect**.

Hydrophobic residues cannot hydrogen bond with the water molecules in an aqueous environment, so instead the water molecules surrounding these residues hydrogen bond with each other in a rigid lattice called a **solvation layer**. These interactions are entropically unfavorable (negative ΔS), and to avoid them, **hydrophobic residues** become **spontaneously buried** in the interior of proteins, away from the aqueous environment.

The passage states that reflectins have few leucine, isoleucine, valine, phenylalanine, or tryptophan residues, all of which are hydrophobic. Instead, reflectins are rich in hydrophilic arginine, serine, and aspartate residues, which facilitate interactions with water molecules. The relative lack of hydrophobic residues means that reflectins experience only a **weak hydrophobic effect** and therefore do not adopt a defined tertiary structure.

(Choices A and C) Reflectins are rich in hydrophilic residues such as arginine and serine, which interact strongly with water through hydrogen bonding.

(Choice D) It is true that reflectins are largely composed of basic arginine residues, but basic proteins fold readily if they have sufficient hydrophobic regions. It is not basicity that prevents reflectins from adopting tertiary structure.

Educational objective:
Proteins frequently fold into specific three-dimensional forms called tertiary structures. The major driving force behind folding is the hydrophobic effect, which causes hydrophobic residues to be buried on the interior of proteins, preventing water from forming solvation layers.

Time spent:0 sec
Subject:
Biochemistry

QID:402381
System:
Amino Acids and Proteins

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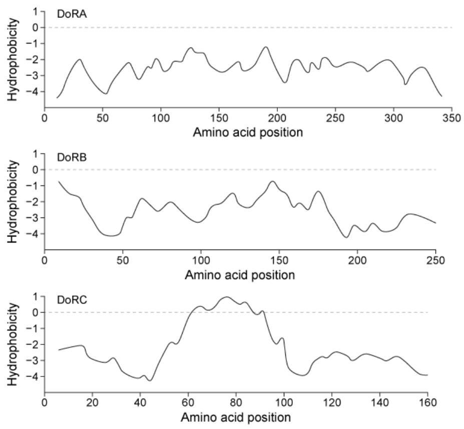

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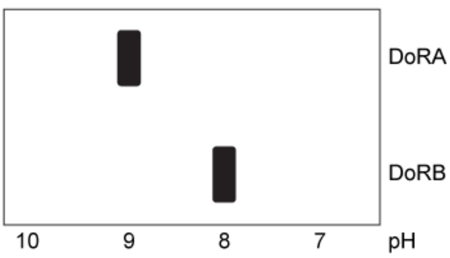


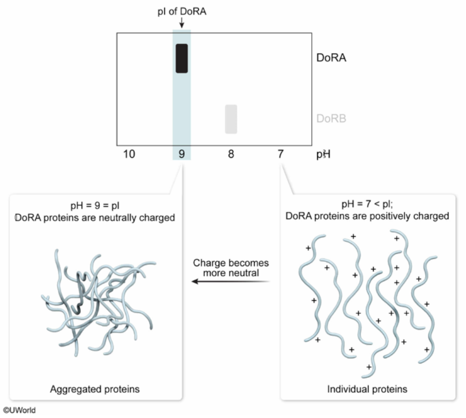
Figure 2 Isoelectric focusing of DoRA and DoRB

If DoRA aggregation is caused by charge neutralization of the protein overall, then aggregates should form as pH:

- ✔ ◉ A. increases from 7 to 9. [60%]
- ○ B. decreases from 7 to 5. [3%]
- ○ C. increases from 9 to 11. [7%]
- ○ D. decreases from 9 to 7. [28%]

Correct
60% answered correctly

Explanation:



The **isoelectric point** (pI) of a protein is the **pH at which the net charge** of the protein is **neutral**. If the ambient **pH is lower than a protein's pI**, the protein gains protons and becomes **positively charged**. If the **pH exceeds the pI**, the protein loses protons and becomes **negatively charged**. When pH = pI, any positive charges in the protein are cancelled by negative charges elsewhere in the protein, resulting in a net charge of zero.

The pI of a protein can be empirically determined by isoelectric focusing, in which proteins migrate through a pH gradient until they are in a pH environment that is equal to their pI. At this point they become **neutrally charged** overall and are **no longer influenced by an electric field**. Therefore, the pH at which a protein stops migrating is its pI.

The scientists in the passage proposed that reflectins aggregate as their net charges are neutralized. If this is correct, then DoRA should have the greatest tendency to aggregate when the ambient pH is equal to its pI. Figure 2 shows that the pI of DoRA is approximately 9, so DoRA should begin to aggregate as the pH **increases from 7 to 9**.

(Choice B) Decreasing the pH from 7 to 5 would move the ambient pH far below the pI of 9 and *increase* the overall positive charge on DoRA. As a result, the tendency of DoRA to aggregate would be greatly *reduced*.

(Choices C and D) DoRA should have the greatest tendency to aggregate at a pH of 9, where it has a neutral charge. Moving the pH away from 9, either by increasing it or decreasing it, would *reduce* the tendency of

DoRA to aggregate.

Educational objective:
The isoelectric point (pI) of a protein is the pH at which the protein is neutrally charged overall and does not migrate in an electric field. When the pH is lower than the pI, the protein becomes protonated and is positively charged. When the pH exceeds the pI, the protein loses protons and becomes negatively charged.

Time spent:0 sec
Subject:
Biochemistry

QID:402382
System:
Amino Acids and Proteins

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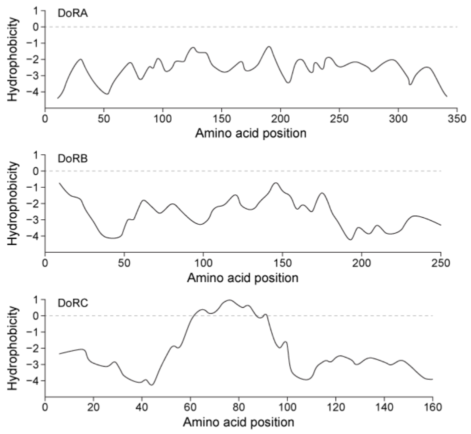

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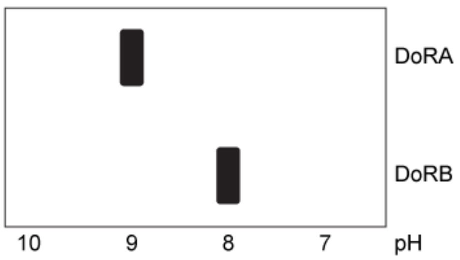


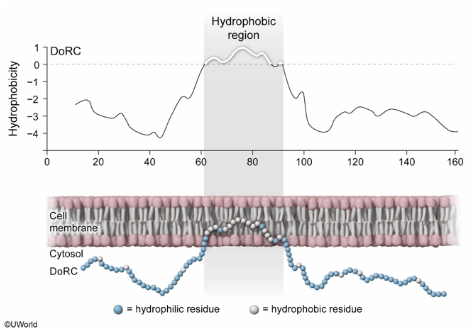
Figure 2 Isoelectric focusing of DoRA and DoRB

Which of the reflectin variants is most likely to be integrated into a membrane?

- A. DoRA only [3%]
- B. DoRB only [6%]
- ✔ C. DoRC only [77%]
- D. Both DoRA and DoRB [12%]

Correct
77% answered correctly

Explanation:



The interiors of **biological membranes** are composed primarily of **hydrophobic** carbon chains. These chains **interact efficiently** with **hydrophobic** amino acid residues but poorly with hydrophilic residues. Conversely, the aqueous environment of the cytosol or the lumen of organelles interacts well with hydrophilic, but not hydrophobic, residues. For this reason, **hydrophobic regions** of proteins are typically found either buried within the protein or **integrated into a membrane** whereas hydrophilic regions are typically exposed to aqueous environments such as cytosol, the lumen of an organelle, or the extracellular space.

Figure 1 shows the hydrophobicity of each reflectin variant as a function of amino acid position. All three reflectin variants are predominantly hydrophilic, but DoRC contains one region between amino acids 60 and 92 that is hydrophobic (hydrophobicity > 0). The passage states that reflectin proteins do *not* adopt a defined tertiary structure, so the hydrophobic region cannot be buried in the interior of the protein. Instead, the hydrophobic region of the **DoRC** protein most likely avoids the formation of highly ordered solvation layers by **integrating into the hydrophobic environment of a membrane**.

(Choices A, B, and D) DoRA and DoRB both lack hydrophobic regions and therefore would interact unfavorably with the hydrophobic environment of a cell membrane.

Educational objective:
Some proteins have regions that are integrated into biological membranes. These regions tend to be hydrophobic, allowing them to interact well with the hydrophobic environment of membrane interiors.

Time spent:0 sec
Subject:
Biochemistry

QID:402383
System:
Amino Acids and Proteins

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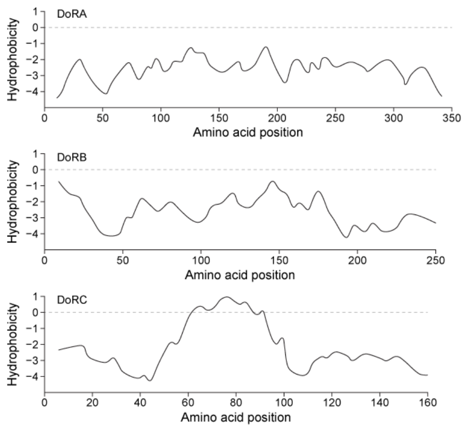

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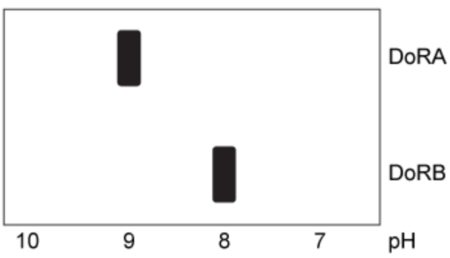


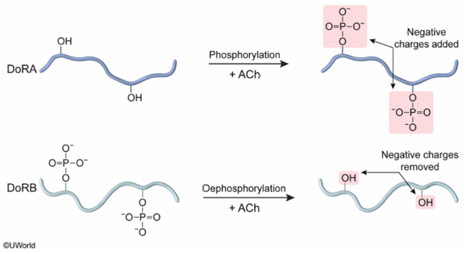
Figure 2 Isoelectric focusing of DoRA and DoRB

What effect does the addition of ACh have on the net charge of DoRA and DoRB in *D. opalascens* skin cells?

- A. The net charges of both DoRA and DoRB become more positive. [8%]
- B. The net charge of DoRA becomes more positive and that of DoRB becomes more negative. [11%]
- ✔ C. The net charge of DoRA becomes more negative and that of DoRB becomes more positive. [70%]
- D. The net charges of both DoRA and DoRB become more negative. [9%]

Correct
71% answered correctly

Explanation:



Post-translational modifications are additions of non-amino acid structures to proteins. Such structures are frequently added and removed to **modulate protein activity**. One of the most common post-translational modifications is **phosphorylation**, in which a phosphate group is attached to a serine, threonine, or tyrosine residue. Phosphates are **negatively charged**, and phosphorylation causes a protein to become more negatively charged overall. In contrast, dephosphorylation makes a protein more positively charged.

The passage states that the reflectin proteins are rich in serine residues, among others, indicating that there are multiple candidate phosphorylation sites. Further, the passage states that in *D. opalescens* skin cells, the **addition of acetylcholine (ACh)** causes a signal cascade that results in **phosphorylation of DoRA** and ***de*phosphorylation of DoRB**.

Therefore, when ACh is added, the **net charge of DoRA** becomes **more negative** because phosphate groups are added, and the **net charge of DoRB** becomes **more positive** (ie, less negative) because phosphate groups are removed.

(Choice A) Only DoRB becomes more positive. DoRA becomes phosphorylated when ACh is added, and therefore becomes more *negatively* charged.

(Choice B) This option would require DoRA to become dephosphorylated and DoRB to become phosphorylated when ACh is added. The passage states that the opposite occurs.

(Choice D) Only DoRA becomes more negative. DoRB becomes *de*phosphorylated when ACh is added and therefore becomes more *positively* charged (ie, less negatively charged).

Educational objective:
Proteins are frequently modified post-translationally with non-amino acid groups such as phosphates. Phosphorylation results in more negative charges on a protein. Dephosphorylation makes a protein more positively charged.

Time spent:0 sec
Subject:
Biochemistry

QID:402384
System:
Amino Acids and Proteins